

INFO 6105

Data Science Engineering Methods and Tools

Northeastern University, Fall 2021

Problem Set 2, Due: Nov 20, 2019

Problem Set Rules:

1. Each student should hand in an individual problem set.
2. Discussing problem sets with other students is permitted. Copying from another person or solution set is not permitted.
3. Late assignments will not be accepted. No exceptions.
4. The solutions should be posted on the website by the beginning of class on the day the problem set is due.
5. How to get data sets: See [Package ISLR](#)

1. (Total: 20 points) We want to use logistic regression to predict the probability of default using income, balance, and student on the [Default data set](#). Our goal is to estimate the test error of this logistic regression model using the validation set approach. Do not forget to set a random seed before beginning your analysis.

(a) (5 points) Using the validation set approach, estimate the test error of a logistic regression model that uses income and balance to predict default. In order to do this, you must perform the following steps:

1. Split the sample set into a training set and a validation set.
2. Fit a multiple logistic regression model using only the training observations.
3. Obtain a prediction of default status for each individual in the validation set by computing the posterior probability of default for that individual, and classifying the individual to the default category if the posterior probability is greater than 0.5.
4. Compute the validation set error, which is the fraction of the observations in the validation set that are misclassified.

```
# Set a random seed
np.random.seed(390)

# Split the data into training and validation sets
train_data, validation_data = train_test_split(data, test_size=0.2, random_state=390)

# Define the features and the target variable
X_train = train_data[['income', 'balance']]
y_train = train_data['default'].apply(lambda x: 1 if x == 'Yes' else 0)

# Fit a logistic regression model on the training set
model = LogisticRegression()
```

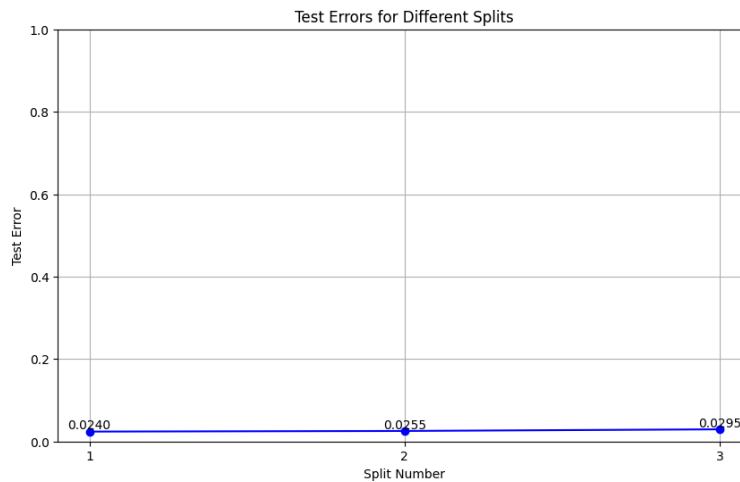
```
model.fit(X_train, y_train)
```

```
# Calculate the test error on the validation set
```

```
test_error = 1 - accuracy_score(y_validation, y_pred)
```

```
Test Error: 0.0240
```

- (b) (7 points) Repeat the process in (a) three times, using three different splits of the observations into a training set and a validation set. Describe your findings and comment on the results obtained.



Test Errors for each split: [0.024, 0.0255, 0.0295]

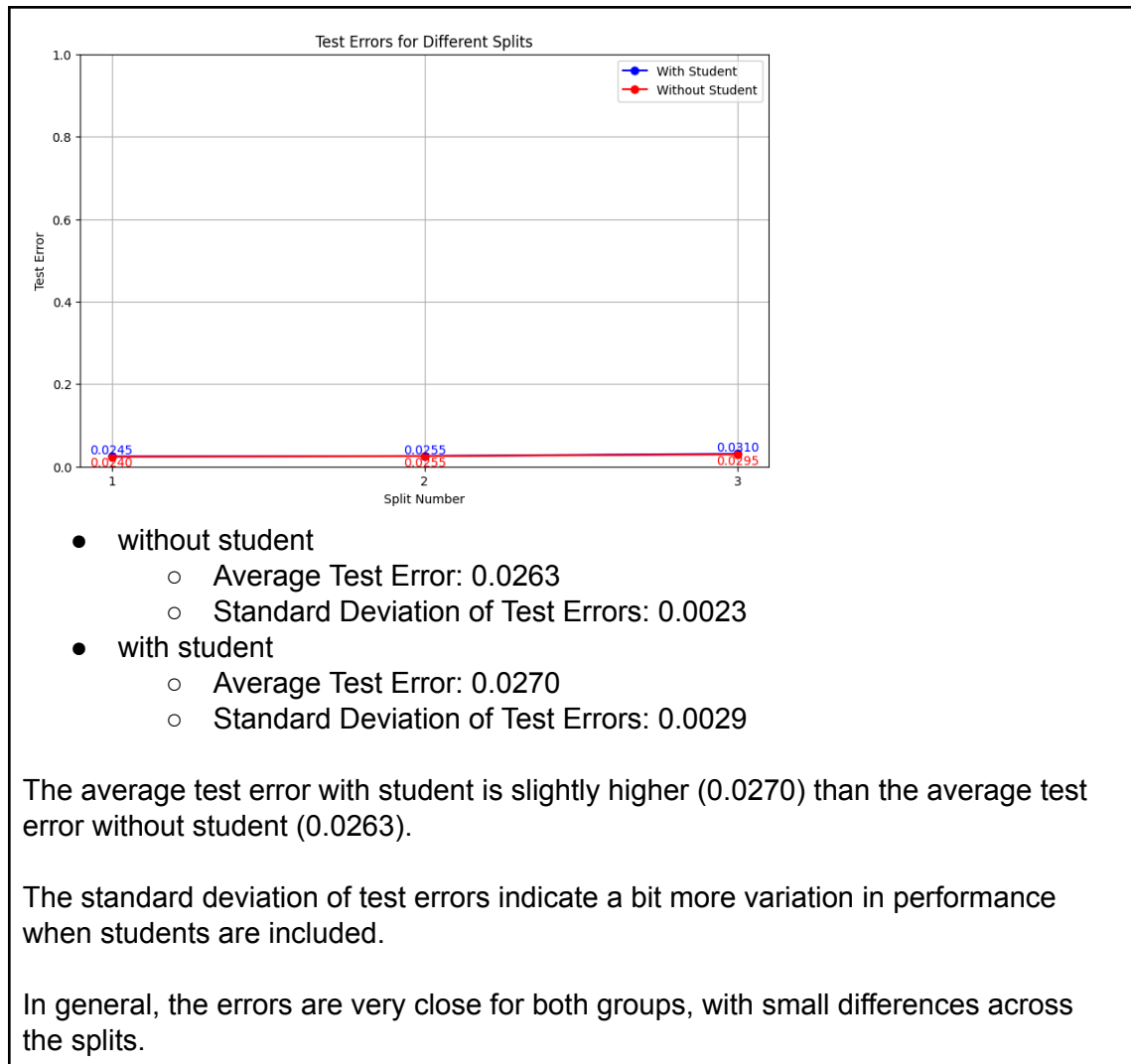
Average Test Error: 0.0263

Standard Deviation of Test Errors: 0.0023

The model's performance is stable with a low mean validation error (0.0263)

The low standard deviation (0.0023) indicates consistent performance across splits.

- (c) (8 points) Now consider a logistic regression model that predicts the probability of default using income, balance, and a dummy variable for student. Estimate the test error for this model using the validation set approach. Comment on whether or not including a dummy variable for student leads to a reduction in the test error rate.



2. (Total: 20 points)

Suppose we collect data for a group of bank customers with variables

- **default** A variable with levels No and Yes indicating whether the customer defaulted on their debt
- **student** A variable with levels No and Yes indicating whether the customer is a student
- **balance** The average balance that the customer has remaining on their credit card after making their monthly payment
- **income** Income of customer

We fit a logistic regression and produce estimated coefficient, $\beta^0 = -15.05$, $\beta^{\text{studentYes}} = -0.5149$, $\beta^{\text{balance}} = 0.00717$, $\beta^{\text{income}} = -0.00000791$.

- (a) (5 points) Estimate the probability that a student with a balance of \$2,000 and income \$45,000 does default on a loan.

The estimated probability that a student with a balance of \$2,000 and income \$45,000 does default on a loan is 0.1707

- (b) (5 points) Estimate the probability that borrower a balance of \$3,000 and income \$70,000 who is not a student does default on a loan.

The estimated probability that a non-student with a balance of \$3,000 and income \$70,000 does default on a loan is 0.9973

- (c) (5 points) How much income would the student in part (a) need to make to have a 90% chance of getting approved for a loan?

The student would need to make an income of \$122,920.35 to have a 90% chance of getting approved for a loan.

- (d) (5 points) How much income would the borrower in part (b) need to make to have a 90% chance of getting approved for a loan?

The non-student borrower would need to make an income of \$1,094,462.71 to have a 90% chance of getting approved for a loan.

3. (Total: 30 points) This problem involves the Boston data set, which can be found in the file Boston.csv. This data set contains the following columns:

crim per capita crime rate by town.

zn proportion of residential land zoned for lots over 25,000 sq.ft.

indus proportion of non-retail business acres per town.

chas Charles River dummy variable (= 1 if tract bounds river; 0 otherwise).

nox nitrogen oxides concentration (parts per 10 million).

rm average number of rooms per dwelling.

age proportion of owner-occupied units built prior to 1940.

dis weighted mean of distances to five Boston employment centres.

rad index of accessibility to radial highways.

tax full-value property-tax rate per \$10,000.

ptratio pupil-teacher ratio by town.

black $1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town.

lstat lower status of the population (percent).

medv median value of owner-occupied homes in \$1,000s.

We want to predict whether a given suburb has a crime rate above or below the median using the other variables in this data set.

- (a) (5 points) For each predictor, fit a simple logistic regression model to predict the response. In which of the models is there a statistically significant association between the predictor and the response (set $\alpha = 0.05$)

Predictors with a statistically significant association ($\alpha = 0.05$):

Predictor	p-value
zn	1.491829e-12
indus	1.004798e-32
chas	1.187845e-01
nox	1.319223e-29
rm	5.605110e-04
age	1.107702e-30
dis	7.574369e-29
rad	4.628257e-09
tax	7.619403e-26
ptratio	3.091749e-08
black	2.252362e-07
lstat	5.879537e-20
medv	2.004576e-08

- All predictors except chas (p-value = 1.187845e-01) show a statistically significant association with the response at $\alpha = 0.05$.
- Significant predictors include zn, indus, nox, rm, age, dis, rad, tax, ptratio, black, lstat, and medv.

- (b) (12 points) Fit a multiple logistic model to predict the response using all of the predictors. Describe your results.
- (i) Do any of the predictors appear to be statistically significant (set $\alpha = 0.05$)? If so, which ones?

Logit Regression Results							
Dep. Variable:	crim_above_median	No. Observations:	506				
Model:	Logit	Df Residuals:	492				
Method:	MLE	Df Model:	13				
Date:	Tue, 03 Dec 2024	Pseudo R-squ.:	0.6979				
Time:	18:34:18	Log-Likelihood:	-105.96				
converged:	True	LL-Null:	-350.73				
Covariance Type:	nonrobust	LLR p-value:	2.438e-96				
	coef	std err	z	P> z	[0.025	0.975]	
const	-34.1037	6.530	-5.223	0.000	-46.902	-21.305	
zn	-0.0799	0.034	-2.369	0.018	-0.146	-0.014	
indus	-0.0594	0.044	-1.358	0.174	-0.145	0.026	
chas	0.7853	0.729	1.077	0.281	-0.643	2.214	
nox	48.5238	7.397	6.560	0.000	34.027	63.021	
rm	-0.4256	0.701	-0.607	0.544	-1.800	0.949	
age	0.0222	0.012	1.814	0.070	-0.002	0.046	
dis	0.6914	0.218	3.167	0.002	0.264	1.119	
rad	0.6565	0.152	4.306	0.000	0.358	0.955	
tax	-0.0064	0.003	-2.385	0.017	-0.012	-0.001	
ptratio	0.3687	0.122	3.019	0.003	0.129	0.608	
black	-0.0135	0.007	-2.069	0.039	-0.026	-0.001	
lstat	0.0439	0.049	0.895	0.371	-0.052	0.140	
medv	0.1671	0.067	2.497	0.013	0.036	0.298	

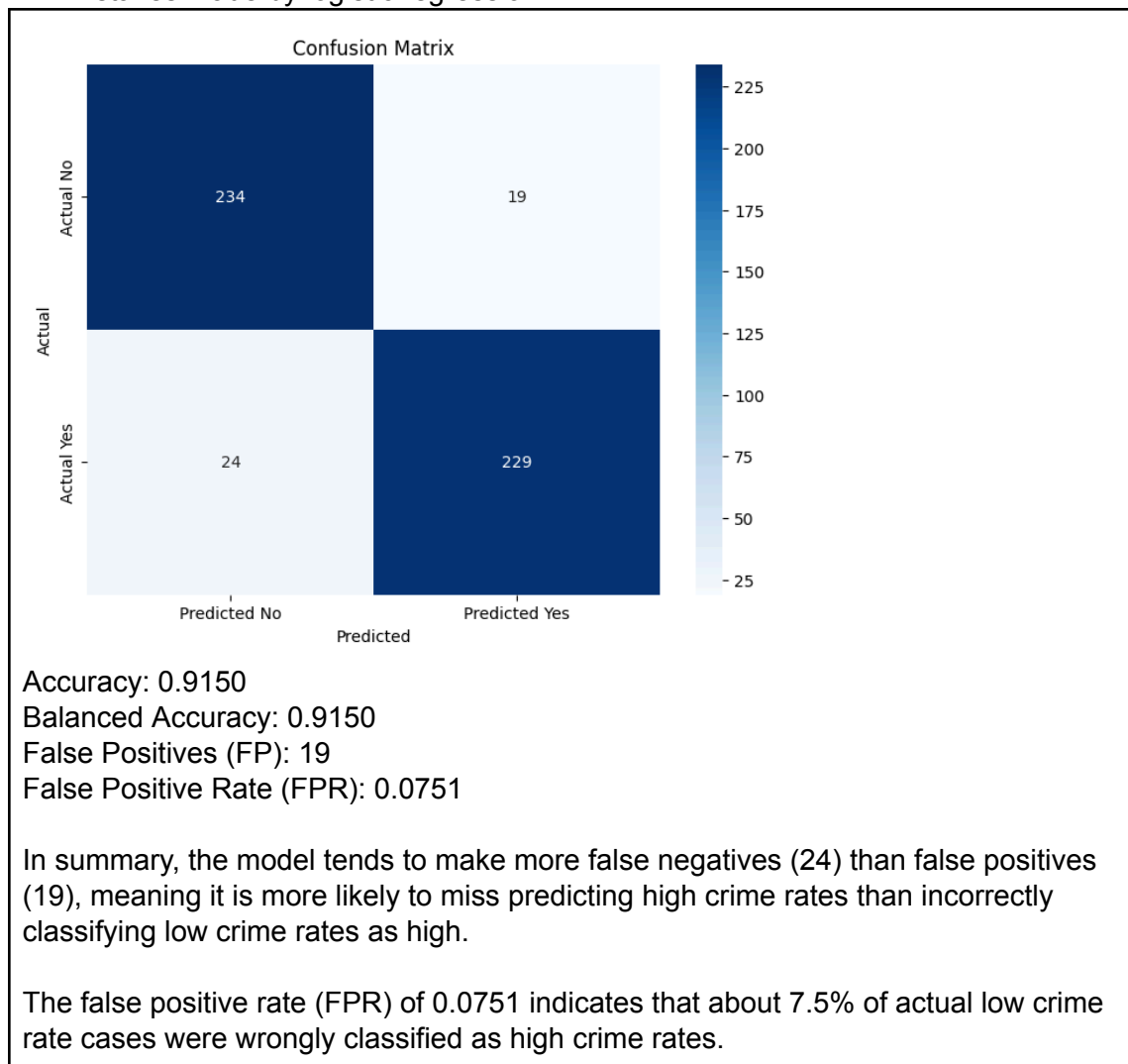
Significant predictors in the multiple logistic regression model ($\alpha = 0.05$):
 ['zn', 'nox', 'dis', 'rad', 'tax', 'ptratio', 'black', 'medv']

- (ii) Predict whether a given suburb has a crime rate above or below the median for a suburb with:

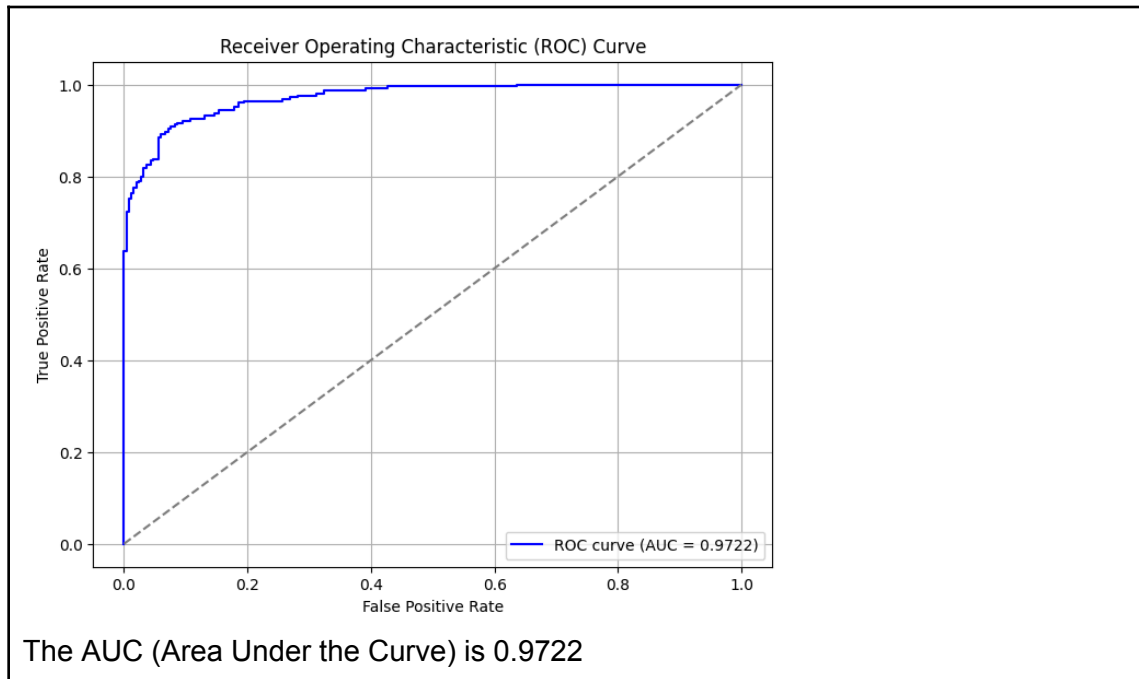
zn=18 indus=2.97 chas= 0 nox= 0.4 rm=6.575 age=63 dis=4.8
 rad=3 tax=238 ptratio=15.3 black=376.7 lstat=8.23 medv=45

The predicted probability that the crime rate is above the median is 0.0040
 The suburb is classified as having a crime rate below the median.

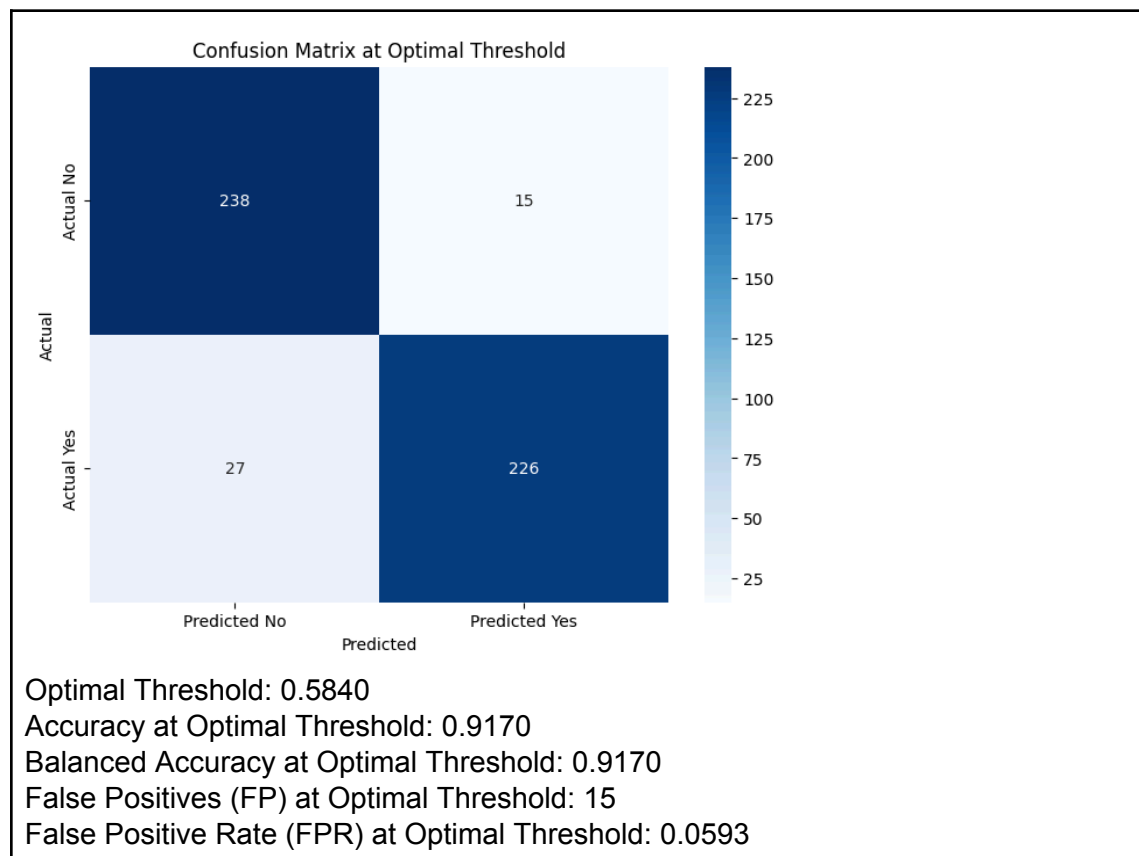
- (iii) Compute the confusion matrix, accuracy, balanced accuracy, FP, and FP rates for Threshold=0.5. Explain what the confusion matrix is telling you about the types of mistakes made by logistic regression.



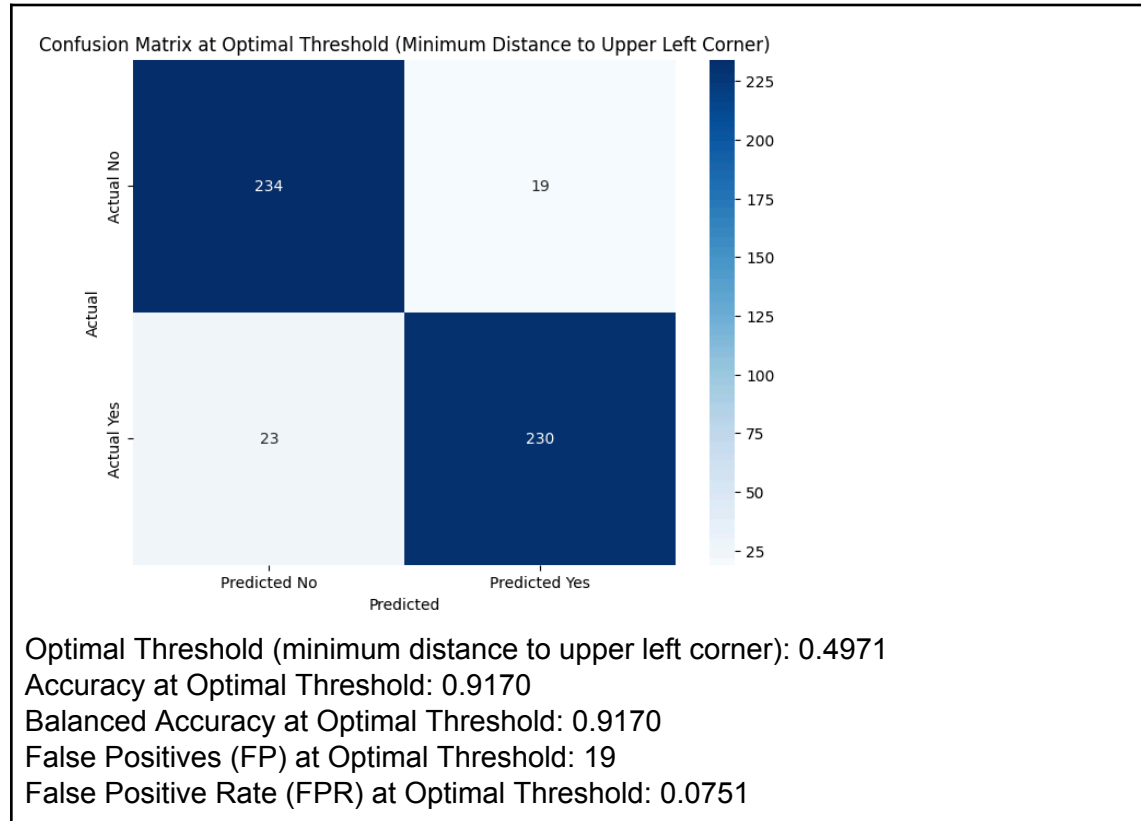
(iv) Draw the ROC curve. What is the AUC?



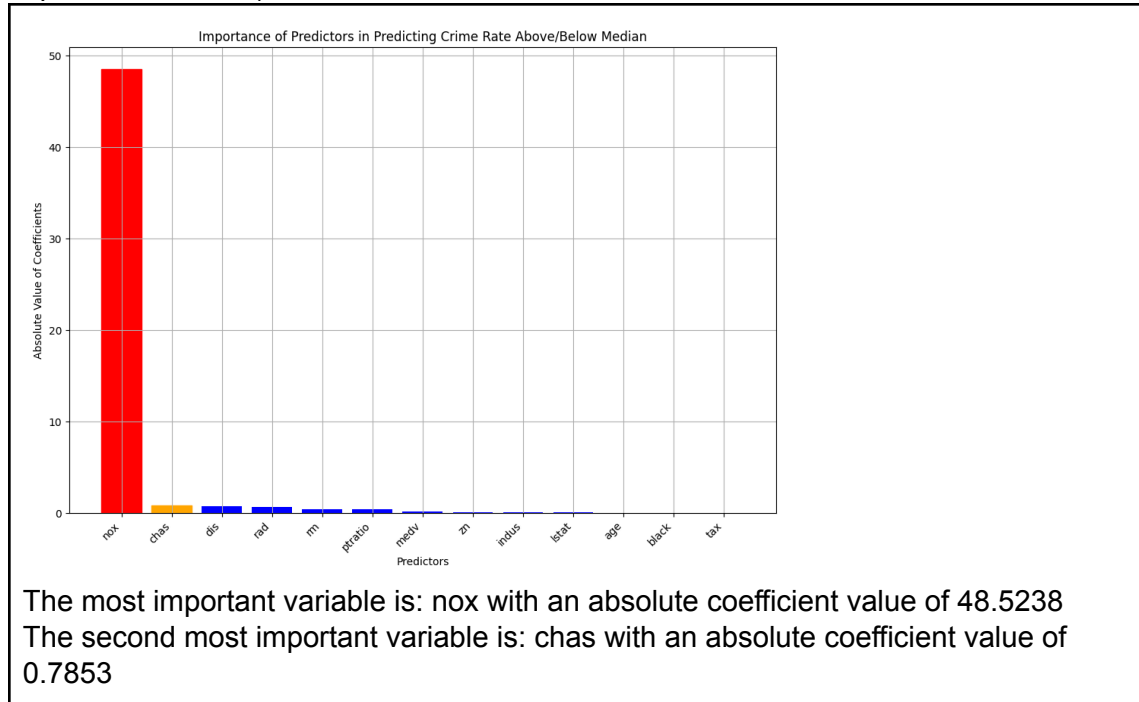
(v) Determine the threshold for which $(TP \text{ rate} + (1 - FP \text{ rate}))$ is maximal. Then, compute the confusion matrix, accuracy, balanced accuracy, FP, and FP rates for that threshold.



- (vi) Determine the threshold for which the ROC curve has the minimum distance to the upper left corner (where TP rate=1 and FP rate=0). Note that this distance is $\sqrt{(1 - \text{TP rate})^2 + (\text{FP rate})^2}$. Then, compute the confusion matrix, accuracy, balanced accuracy, FP, and FP rates for that threshold.



- (c) (6 points) Which predictors matter most for predicting whether a given suburb has a crime rate above or below the median? (Find the first and the second most important variables)



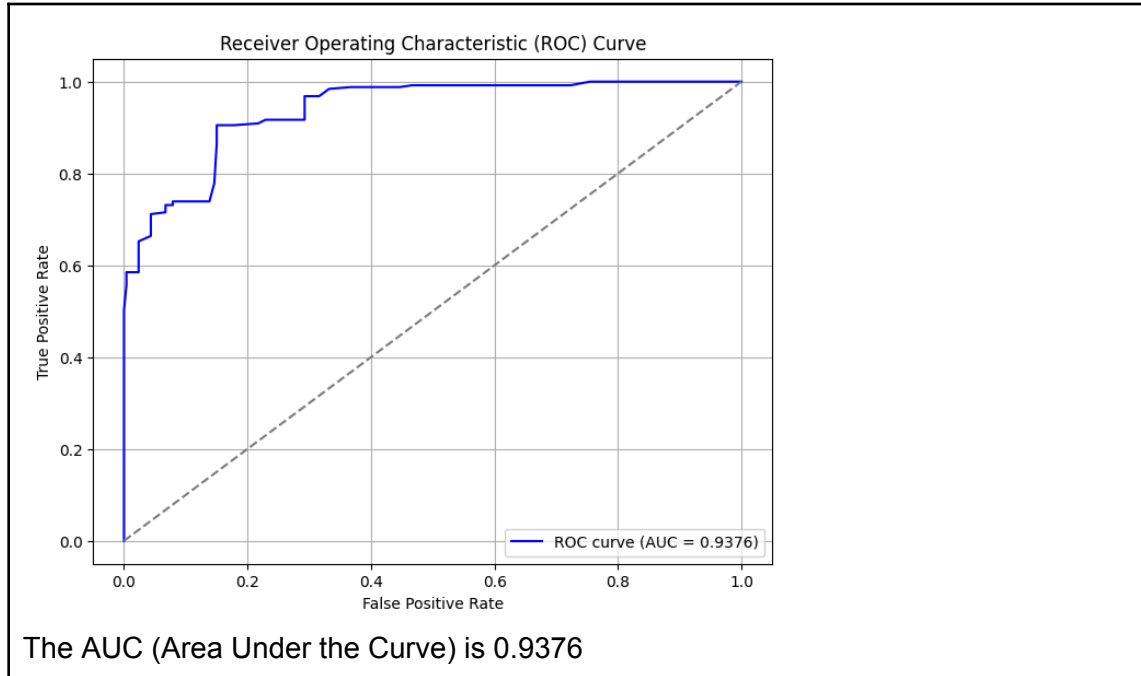
(d) (7 points) Fit a multiple logistic model to predict the response using the first two most important variables in Part (c).

(i) Write out the model in equation form.

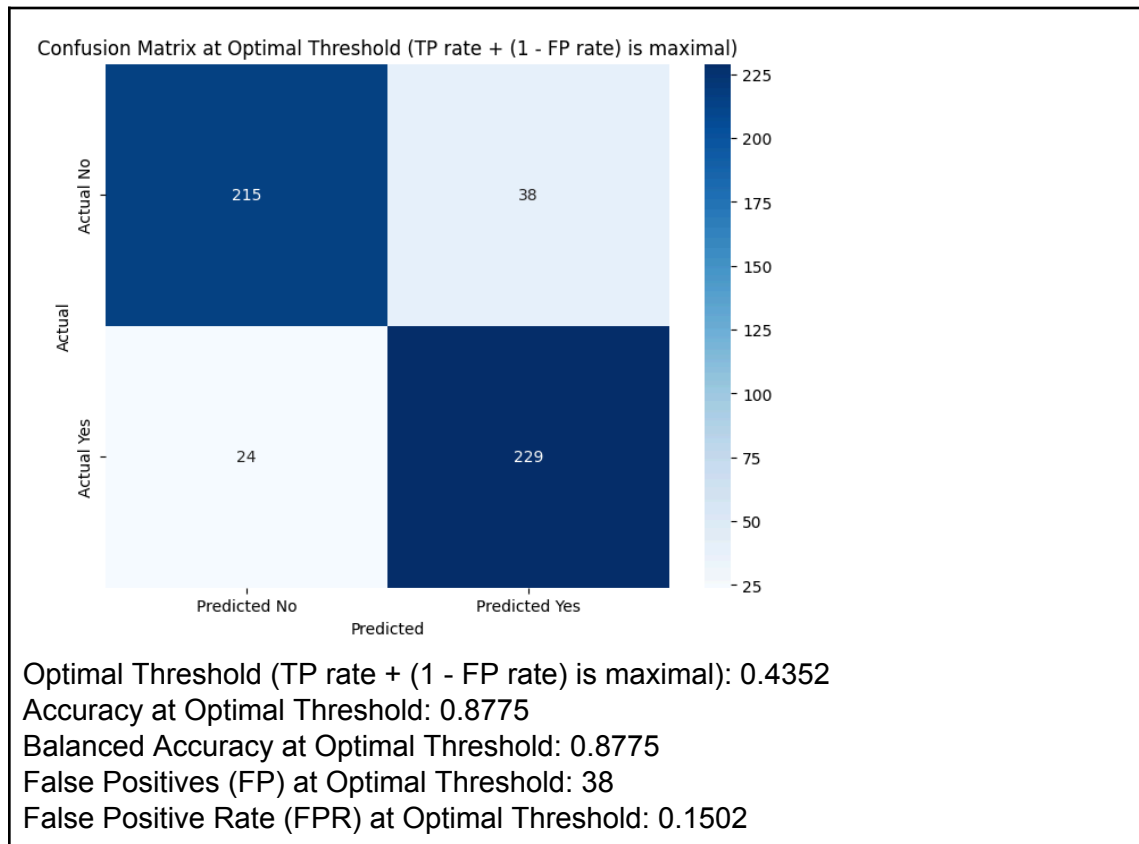
Logistic Regression Model Equation:

$$\text{logit}(p) = -16.1118 + 29.7958 * \text{nox} + 0.8756 * \text{chas}$$

(ii) Draw the ROC curve. What is the AUC?



- (iii) Determine the threshold for which (TP rate + (1-FP rate)) is maximal. Then, compute the confusion matrix, accuracy, balanced accuracy, FP, and FP rates for that threshold.



4. (Total: 8 points) Suppose we produce ten bootstrapped samples from a data set containing red and green classes. We then apply a classification tree to each bootstrapped sample and, for a specific value of X , produce 10 estimates of $P(\text{Class is Red} \mid X)$:

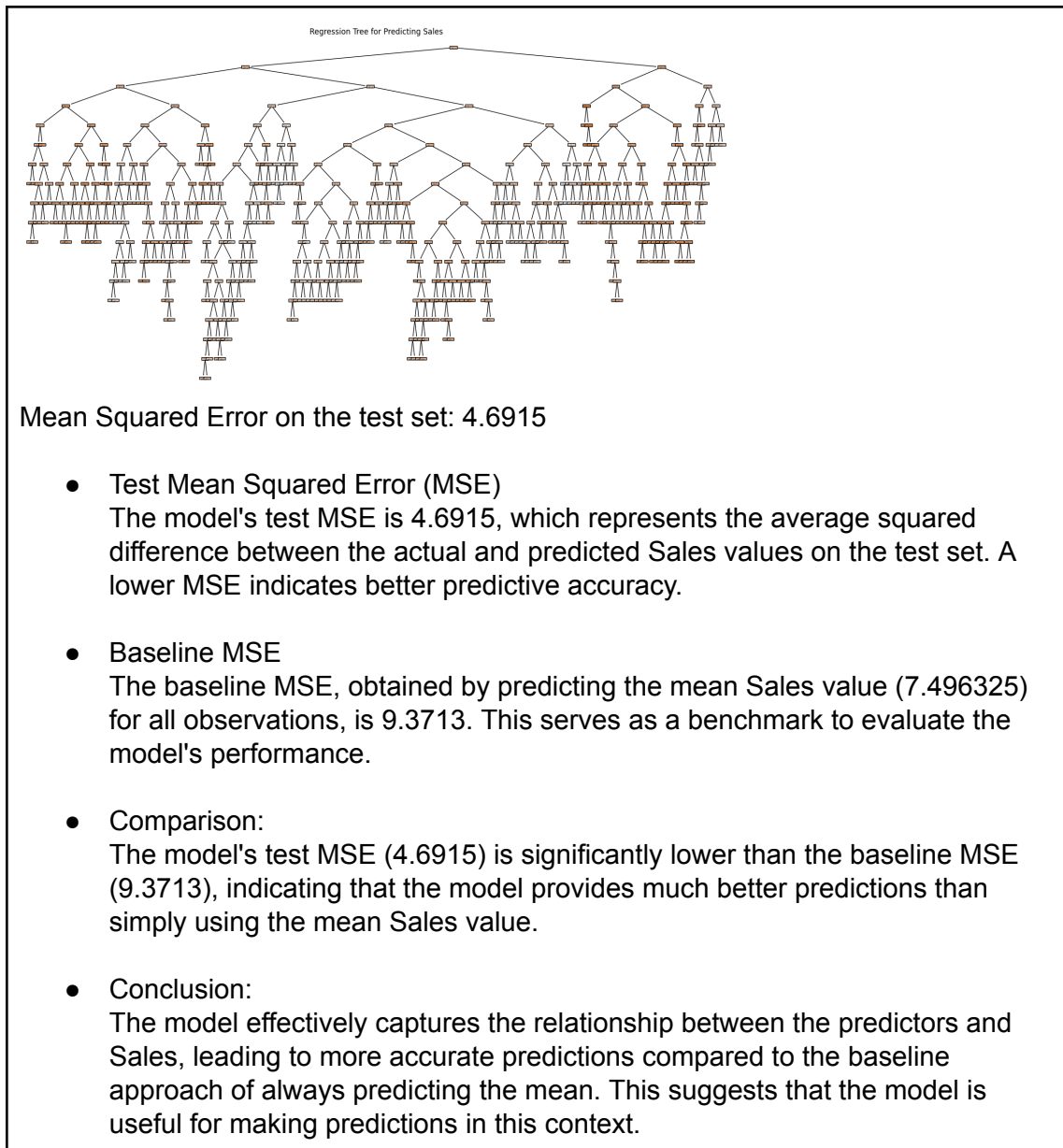
0.1, 0.15, 0.2, 0.2, 0.55, 0.6, 0.6, 0.65, 0.7, and 0.75.

There are two common ways to combine these results together into a single class prediction. One is the majority vote approach. The second approach is to classify based on the average probability. In this example, what is the final classification under each of these two approaches?

Final classification under the majority vote approach: Red

Final classification under the average probability approach: Not Red

5. (Total: 26 points) This problem involves the Carseats data set. We want to predict Sales using regression trees. Split the data set into a training set (75%) and a test set (0.25%).
- (a) (6 points) Fit a regression tree to the training set. Plot the tree, and interpret the results. What test MSE do you obtain?



- (b) (6 points) Use cross-validation in order to determine the optimal level of tree complexity. Does pruning the tree improve the test MSE?

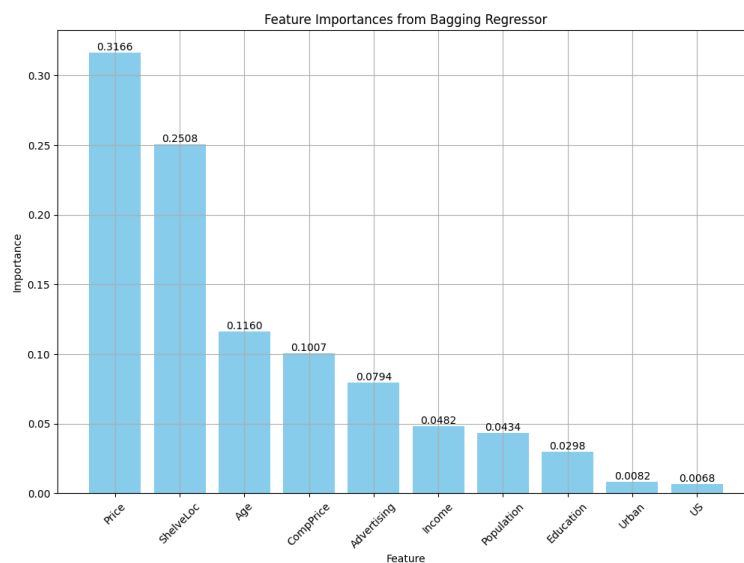
Best max_depth: 12

Test MSE with the best model: 4.5046

Test MSE with the unpruned tree: 4.6915

Pruning the tree to a maximum depth of 12 has improved the test MSE from 4.6915 to 4.5046. This indicates that the pruned tree generalizes better to the test data compared to the unpruned tree.

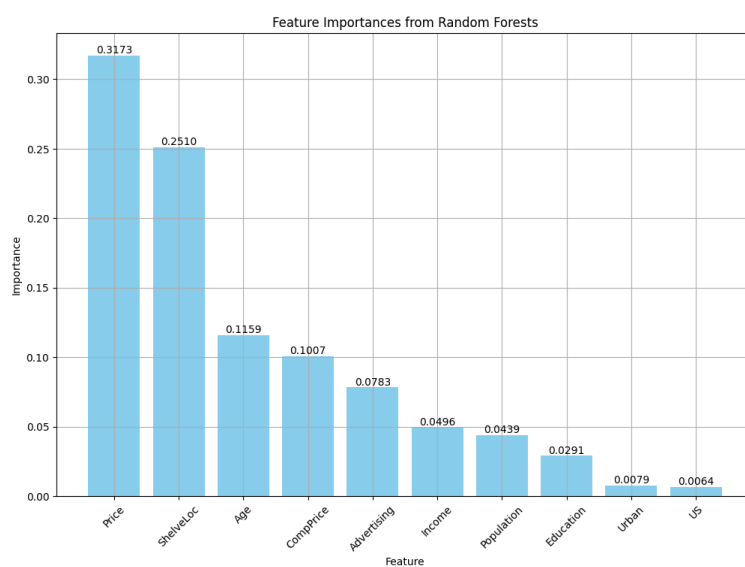
- (c) (7 points) Use the bagging approach in order to analyze this data. What test MSE do you obtain? Determine which variables are most important.



Test MSE with bagging: 2.7287

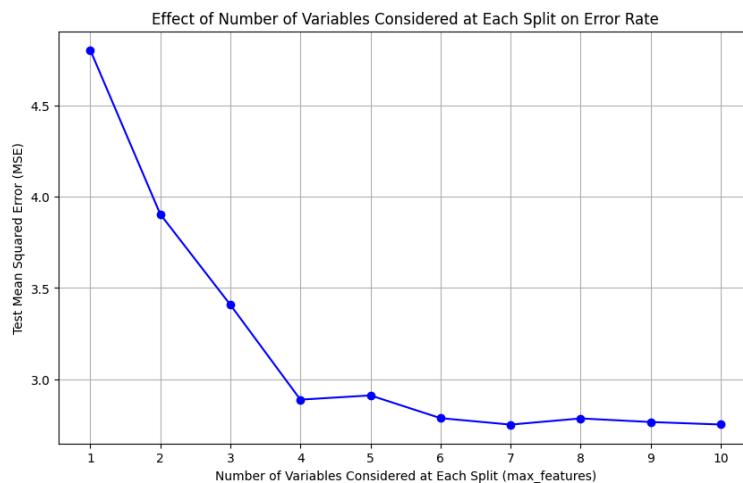
- Test Mean Squared Error (MSE): The test MSE using the bagging approach is 2.7287, significantly lower than the baseline MSE (9.3713) and the previously reported test MSE (4.6915). This indicates that bagging improves the model's predictive performance.
- Feature Importance: The most important variables based on their importance scores are:
 - Price (importance = 0.3166)
 - ShelfLoc (importance = 0.2508)
 - Age (importance = 0.1160)

- (d) (7 points) Use random forests to analyze this data. What test MSE do you obtain? Determine which variables are most important. Describe the effect of the number of variables considered at each split on the error rate obtained.



Test MSE with random forests: 2.7520

- Test Mean Squared Error (MSE): The test MSE with random forests is 2.7520, which is slightly higher than the test MSE from bagging (2.7287) but still significantly lower than the baseline MSE (9.3713). This indicates strong predictive performance.
- Feature Importance: The most important variables based on their importances are:
 - Price (importance = 0.3173)
 - ShelfLoc (importance = 0.2510)
 - Age (importance = 0.1159)

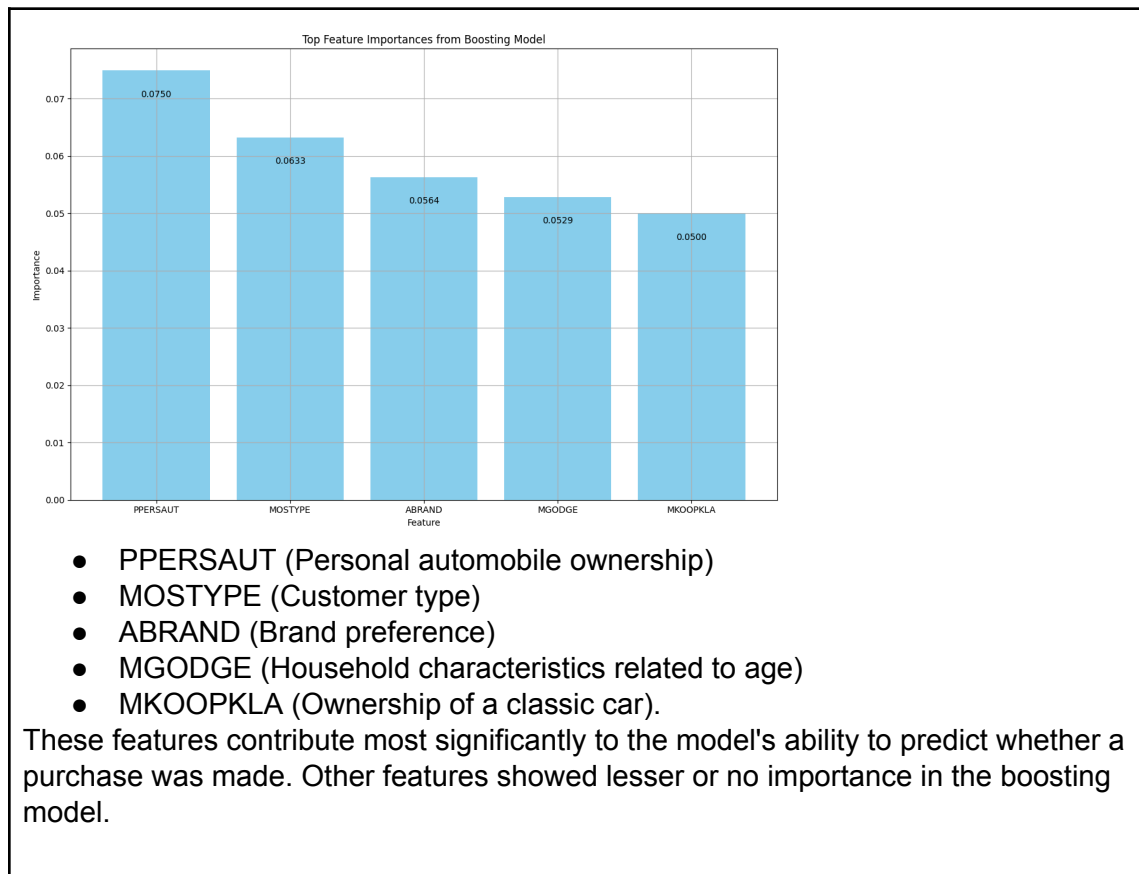


- Initial Improvement: Increasing max_features from 1 to 4 reduces the test MSE significantly (from 4.8031 to 2.8881), indicating improved splits and better performance.

- Optimal Range: The lowest test MSE (2.7514) occurs at max_features = 7, suggesting this is the optimal number of variables for this dataset.
- Stabilization: Beyond max_features = 7, the test MSE stabilizes, ranging from 2.7514 to 2.9113, with no significant performance gains.
- The optimal number of variables at each split is around 7. While initial increases in max_features improve performance, further increases beyond this point offer diminishing returns.

6. (Total: 26 points) This question uses the Caravan data set. Create a training set consisting of the first 1,000 observations, and a test set consisting of the remaining observations.

(a) (10 points) Fit a boosting model to the training set with Purchase as the response and the other variables as predictors. Use 1,000 trees, and a shrinkage value of 0.01. Which predictors appear to be the most important?



(b) (16 points) Use the boosting model to predict the response on the test data. Predict that a person will make a purchase if the estimated probability of purchase is greater than 20 %. Form a confusion matrix. What fraction of the people predicted to make a purchase do in fact make one? How does this compare with the results obtained from applying logistic regression to this data set?

